Fengwei Fried Rice · Stage 2 Test Bank

For the study guide "Hot Wok, Smart Chips" · Hardware, Networks & Behaviourist AI · November 2026

Contents: 25 True/False, 25 multiple-choice (21 choose-one, 4 select-all) and 10 open questions, 60 in total, numbered 1-60. Each question is tagged [Recipe · guide section · level]. Levels: R = remember, U = understand, A = apply, An = analyse, E = evaluate, C = create, P = predict. Questions tagged "Lab" use a Kitchen Lab from the guide; all the data you need is in the question, so no website or app is required. The answer key starts on a new page after all questions.

Coverage

Topic (guide recipe)	True/False	Multiple choice	Open
Recipe 01 · AI Hardware	5	5	2
Recipe 02 · Networks & AI Infrastructure	5	5	2
Recipe 03 · Reinforcement Learning	5	5	2
Recipe 04 · AI Sensors	5	5	2
Recipe 05 · System Integration	5	5	2
Total	25	25	10

Part A · True or False

Write T (true) or F (false) for each statement.

1. One MAC (multiply-accumulate) counts as two FLOPs. *[R1 · 1.1 · R]*

2. A TPU is more flexible than a CPU for general programs. *[R1 · 1.2 · U]*

3. If 90% of a job can be parallelised, 100 cooks give a speed-up of 100×. *[R1 · 1.3 · A]*

4. At batch size 1, a large layer on a typical accelerator is usually memory-bound. *[R1 · 1.4 · U]*

5. In Kitchen Lab 1 (Race the Crew), the GPU's speed-up over the CPU is LARGER for 256 × 256 matrices than for 4,096 × 4,096 matrices. *[R1 · Lab 1 · A]*

6. Encapsulation means each layer adds its own header to the data from the layer above. *[R2 · 2.1 · U]*

7. UDP guarantees that every packet arrives in order. *[R2 · 2.2 · U]*

8. Bandwidth and latency are two names for the same quantity. *[R2 · 2.4 · U]*

9. In a ring all-reduce, the traffic per chip approaches 2 × S as the number of chips grows. *[R2 · 2.4 · A]*

10. In Kitchen Lab 2, a server answers ping with an average round-trip time of 12 ms. Using the speed-of-light floor of 5 µs per km (one way), the server can be at most about 1,200 km away. *[R2 · Lab 2 · A]*

11. With rewards [0, 0, 0, 10] and γ = 0.9, the discounted return is 7.29. *[R3 · 3.2 · A]*

12. A discount factor γ close to 0 makes the agent far-sighted. *[R3 · 3.2 · U]*

13. Reward hacking means the agent earns a high reward without doing the intended task. *[R3 · 3.3 · U]*

14. A purely greedy agent can get stuck on a poor option forever. *[R3 · 3.4 · U]*

15. In Kitchen Lab 3 (GridWorld), setting epsilon = 0 guarantees that the agent eventually finds the best route. *[R3 · Lab 3 · U]*

16. A single camera directly measures the distance to every pixel. *[R4 · 4.1 · R]*

17. A LiDAR pulse returning after 400 ns means the object is about 60 m away. *[R4 · 4.1 · A]*

18. Radar measures the speed of an object directly using the Doppler effect. *[R4 · 4.1 · U]*

19. Two sensors that fail for the same reason (for example fog) give the same benefit as two independent sensors. *[R4 · 4.4 · U]*

20. In Kitchen Lab 4 (phyphox), a phone lying still on a table shows an acceleration (without g) of exactly 0 m/s². *[R4 · Lab 4 · U]*

21. Fusing three element-wise operations on a 2 MB tensor cuts memory traffic from 12 MB to 4 MB. *[R5 · 5.1 · A]*

22. In data parallelism, each chip stores only a slice of the model's weights. *[R5 · 5.2 · U]*

23. A PUE of 1.0 means the facility uses no energy beyond the computing equipment. *[R5 · 5.4 · U]*

24. If each of 4,096 chips has an MTBF of 43,800 hours, the whole system fails about every 10.7 hours. *[R5 · 5.4 · A]*

25. In Kitchen Lab 5, the FIRST call of a jax.jit function is much slower than later calls mainly because XLA compiles it. *[R5 · Lab 5 · U]*

Part B · Multiple choice

Each question says how many answers to choose: "(Choose ONE)" means exactly one correct option; "(Select ALL that apply)" means one or more options are correct and you must choose all of them.

26. (Choose ONE) Weights [2, −1, 0.5], inputs [1, 4, 6]. What is the neuron output (no bias)? *[R1 · 1.1 · A]*

- A) 1.0
- B) 3
- C) 5
- D) −1

27. (Choose ONE) Which chip best fits a voice assistant in a small battery-powered speaker? *[R1 · 1.2 · U]*

- A) GPU
- B) Server CPU
- C) NPU
- D) TPU pod

28. (Choose ONE) In a 2 × 2 systolic array with W = [[1, 2], [3, 4]] and x = [5, 6], the output y = W·x is: *[R1 · 1.2 · A]*

- A) [12, 30]
- B) [23, 34]
- C) [11, 23]
- D) [17, 39]

29. (Choose ONE) A chip has 10 TFLOP/s peak and 500 GB/s memory bandwidth. Its ridge point is: *[R1 · 1.4 · A]*

- A) 2 FLOP/byte
- B) 20 FLOP/byte
- C) 50 FLOP/byte
- D) 5,000 FLOP/byte

30. (Choose ONE) Kitchen Lab 1: ten 4,096 × 4,096 matrix multiplications (about 687 billion MACs in total) take 0.25 s on a T4 GPU. What throughput did the GPU reach? *[R1 · Lab 1 · A]*

- A) 11 TFLOP/s
- B) 0.07 TFLOP/s
- C) 2.7 TFLOP/s
- D) 5.5 TFLOP/s

31. (Select ALL that apply) Which of these does TCP provide that UDP does NOT? *[R2 · 2.2 · U]*

- A) Acknowledgement of delivered data
- B) In-order delivery
- C) Port numbers to reach the right program
- D) Retransmission of lost packets

32. (Choose ONE) Which protocol is built on UDP to speed up connection set-up for the web? *[R2 · 2.2 · U]*

- A) QUIC
- B) ARP
- C) SMTP
- D) FTP

33. (Choose ONE) A factory camera must stop a machine within 15 ms. The best place to run the model is: *[R2 · 2.3 · U]*

- A) A distant cloud region
- B) On or near the machine (edge)
- C) A user's laptop at home
- D) Any server; latency does not matter

34. (Choose ONE) Tail latency refers to: *[R2 · 2.4 · U]*

- A) Upload speed
- B) The average delay
- C) The slowest small fraction of requests
- D) The delay of the last packet only

35. (Choose ONE) Multi-access edge computing (MEC) in 5G mainly helps AI services by: *[R2 · 2.5 · U]*

- A) Putting computing close to the base station
- B) Replacing TCP
- C) Removing the need for antennas
- D) Increasing screen resolution

36. (Choose ONE) A corridor of squares 1-2-3 leads to the goal, one step after square 3 (reward 1). With $\gamma = 0.9$, $V(1)$ equals: *[R3 · 3.2 · A]*

- A) 1.0
- B) 0.729
- C) 0.81
- D) 0.9

37. (Choose ONE) Q = 0.2, r = 1, max Q(s′) = 0.5, α = 0.1, γ = 0.9. After one Q-learning update the new Q is: *[R3 · 3.3 · A]*

- A) 0.45
- B) 1.45
- C) 0.245
- D) 0.325

38. (Select ALL that apply) Which of these rewards are DENSE (given at almost every step)? *[R3 · 3.3 · U]*

- A) +speed every step
- B) +1 only on reaching the charging station
- C) −energy used every step
- D) +1 only for winning the game

39. (Choose ONE) A bandit has arms that pay 0.2, 0.5 and 0.8 on average. The agent knows the best arm and uses ε = 0.1 (when exploring, it picks one of the three arms at random). Expected reward per pull: *[R3 · 3.4 · A]*

- A) 0.50
- B) 0.77
- C) 0.80
- D) 0.74

40. (Choose ONE) Domain randomisation helps sim-to-real transfer by: *[R3 · 3.5 · U]*

- A) Removing the reward
- B) Using only real data
- C) Making the simulator perfect
- D) Training on many varied simulated conditions

41. (Select ALL that apply) Which sensors measure distance DIRECTLY? *[R4 · 4.1 · U]*

- A) LiDAR
- B) A single camera
- C) Radar
- D) Ultrasonic

42. (Choose ONE) At 100 km/h, how far does a car travel during 0.3 s of processing latency? *[R4 · 4.2 · A]*

- A) 8.33 m
- B) 30 m
- C) 64.3 m
- D) 2.78 m

43. (Choose ONE) Radar says 50.0 m (σ = 0.5 m) and a camera says 52.0 m (σ = 2.0 m). The inverse-variance fused estimate is about: *[R4 · 4.3 · A]*

- A) 52.0 m
- B) 50.12 m
- C) 51.0 m
- D) 51.5 m

44. (Choose ONE) A stereo camera has f = 1,000 px and baseline B = 0.3 m. A disparity of 6 px means a depth of: *[R4 · 4.3 · A]*

- A) 300 m
- B) 42.9 m
- C) 50 m
- D) 60 m

45. (Choose ONE) Under SAE J3016, a driver-assist that steers and brakes but needs constant human supervision is: *[R4 · 4.4 · U]*

- A) Level 2
- B) Level 4
- C) Level 5
- D) Level 0

46. (Choose ONE) Training a 7B-parameter model with mixed-precision Adam needs about 16 bytes per parameter. Total training state: *[R5 · 5.2 · A]*

- A) 14 GB
- B) 56 GB
- C) 112 GB
- D) 224 GB

47. (Choose ONE) A 7 GB INT8 model is served on a chip with 1 TB/s memory bandwidth. The memory-bound limit for one user is about: *[R5 · 5.3 · A]*

- A) 1,000 tokens/s
- B) 4,571 tokens/s
- C) 14 tokens/s
- D) 143 tokens/s

48. (Choose ONE) The diameter (longest shortest path) of a 16 × 16 × 16 torus is: *[R5 · 5.5 · A]*

- A) 16 hops
- B) 24 hops
- C) 45 hops
- D) 48 hops

49. (Select ALL that apply) Which changes usually RAISE the MFU (model FLOPs utilisation) of a training job? *[R5 · 5.4 · U]*

- A) Fusing operations to cut memory traffic
- B) Overlapping communication with computation
- C) Making every step wait for a slow data loader
- D) Using a larger batch per chip

50. (Choose ONE) Kitchen Lab 5: you change the matrix size in the JAX program from 4,096 to 8,192. By what factor do the FLOPs of a @ a grow? *[R5 · Lab 5 · A]*

- A) 16×
- B) 2×
- C) 4×
- D) 8×

Part C · Open questions

Write a developed answer of more than 100 words, or show full working (calculations, a diagram or worked steps) with a short explanation of what the result means.

51. A layer has 512 inputs and 2,048 outputs, with FP16 weights, on a chip with 10 TFLOP/s and 500 GB/s. Compute the arithmetic intensity at batch 1 and at batch 32, decide whether each is memory-bound or compute-bound, estimate the achieved TFLOP/s, and explain in your own words why batching helps. *[R1 · 1.4 · A]*

52. Evaluate running Face ID on the phone's Neural Engine instead of in the cloud. Give two advantages and two limitations of on-device AI, and decide whether the same choice would suit a large chatbot model. Justify your decision. *[R1 · 1.5 · E]*

53. A hospital wants an AI that reads X-rays in under 2 seconds and keeps patient data private, but the model needs a large GPU. Recommend cloud, edge (on-premises) or hybrid, and justify the design with latency, bandwidth, privacy and cost. *[R2 · 2.3 · E]*

54. 14 GB of gradients are all-reduced across 256 chips with 100 GB/s per chip, and one compute step takes 8.2 s. Calculate the traffic per chip, the communication time and the scaling efficiency, then explain one way to improve it and why it works. *[R2 · 2.4 · A]*

55. Design a reward for a home cleaning robot. Show how a naive reward could be hacked, propose a better reward, say whether it is sparse or dense, and compare the two rewards: which behaviour would each one produce? *[R3 · 3.3 · C]*

56. A student runs Kitchen Lab 3 (GridWorld TD learning) with epsilon = 0. After 2,000 steps the agent still walks the same long route; the arrows on squares it never visited still show their starting values. Explain why this plan failed, what the student should change, and what they should expect with epsilon = 0.5. Connect the result to the

greedy agent in the bandit simulation and compare ε-greedy with UCB as a fix. *[R3 · Lab 3 · An]*

57. A car at 50 km/h has 0.15 s sensing-and-processing latency and brakes at 7 m/s². Compute the total stopping distance and the impact speed if a ball is detected 15 m ahead. What does this show about sensor range and latency when you design a safety system? *[R4 · 4.4 · A]*

58. In the offline version of Kitchen Lab 4, two students stand 20 m apart. One claps; the other starts a stopwatch on SEEING the clap and stops it on HEARING it. Five trials give t = 0.21, 0.08, 0.15, 0.30 and 0.11 s. Compute the speed of sound from each trial and from the mean time, compare with 343 m/s, and explain why single results scatter so much. Why does averaging help, what can averaging NOT fix, and how would you improve the experiment? *[R4 · Lab 4 · An]*

59. Estimate the training time, cost and energy for a 7B model on 1.4 trillion tokens with 256 chips (200 TFLOP/s each, 40% MFU, $2 per chip-hour, 400 W per chip, PUE 1.1). Use FLOPs ≈ 6 × parameters × tokens. Which single change would cut all three numbers, and why? *[R5 · 5.4 · A]*

60. Kitchen Lab 5 runs this JAX program on a Colab T4 GPU (FP32 peak about 8.1 TFLOP/s):

```
f = jax.jit(lambda a: (a @ a).sum())
```

Measured times: size 1,024: first call 0.62 s, second call 0.0021 s; size 4,096: 0.034 s; size 8,192: 0.27 s. Explain the slow first call, compute the TFLOP/s and MFU for each size, and explain why the time ratio from 1,024 to 4,096 is far from the FLOP ratio while 4,096 → 8,192 is close to 8×. Predict the time for size 16,384. *[R5 · Lab 5 · P]*

Answer Key

Part A · True or False (false statements are corrected)

1. True. One multiply plus one add = 2 FLOPs. [R1 · 1.1 · R]

2. False. Correct: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work. [R1 · 1.2 · U]

3. False. Correct: Amdahl's law gives 1 ÷ (0.1 + 0.9/100) ≈ 9.2×; the limit is 10×, set by the serial 10%. [R1 · 1.3 · A]

4. True. Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte. [R1 · 1.4 · U]

5. False. Correct: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster. [R1 · Lab 1 · A]

6. True. Like boxes inside boxes: application data → TCP/UDP header → IP header → link header. [R2 · 2.1 · U]

7. False. Correct: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering. [R2 · 2.2 · U]

8. False. Correct: bandwidth is how much data per second; latency is how long one message takes to arrive. [R2 · 2.4 · U]

9. True. Traffic = $2(N - 1)/N \times S$, which tends to 2S for large N. [R2 · 2.4 · A]

10. True. One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer. [R2 · Lab 2 · A]

11. True. $0.9^3 \times 10 = 7.29$. [R3 · 3.2 · A]

12. False. Correct: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted. [R3 · 3.2 · U]

13. True. Example: the boat-racing agent circling to hit respawning targets instead of finishing the race. [R3 · 3.3 · U]

14. True. It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation). [R3 · 3.4 · U]

15. False. Correct: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route. [R3 · Lab 3 · U]

16. False. Correct: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly. [R4 · 4.1 · R]

17. True. $d = c \times t \div 2 = 3 \times 10^8 \times 4 \times 10^{-7} \div 2 = 60$ m. [R4 · 4.1 · A]

18. True. The frequency shift of the echo gives the relative speed in one measurement. [R4 · 4.1 · U]

19. False. Correct: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors. [R4 · 4.4 · U]

20. False. Correct: the reading jitters around 0 (about ±0.05 m/s²) and is never exactly zero; that jitter is sensor noise σ. [R4 · Lab 4 · U]

21. True. Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB. [R5 · 5.1 · A]

22. False. Correct: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism. [R5 · 5.2 · U]

23. True. PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion. [R5 · 5.4 · U]

24. True. 43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints. [R5 · 5.4 · A]

25. True. The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds. [R5 · Lab 5 · U]

Part B · Multiple choice

26. A) 1.0. $2 \times 1 + (-1) \times 4 + 0.5 \times 6 = 2 - 4 + 3 = 1.0$. [R1 · 1.1 · A]

27. C) NPU. An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power. [R1 · 1.2 · U]

28. D) [17, 39]. $1 \cdot 5 + 2 \cdot 6 = 17$; $3 \cdot 5 + 4 \cdot 6 = 39$. [R1 · 1.2 · A]

29. B) 20 FLOP/byte. $10 \times 10^{12} \div (500 \times 10^9) = 20$ FLOP/byte; below 20 the chip is memory-bound. [R1 · 1.4 · A]

30. D) 5.5 TFLOP/s. TFLOP/s = 2 × MACs ÷ time = $2 \times 6.87 \times 10^{11} \div 0.25 \approx 5.5 \times 10^{12}$. 2.7 forgets the factor 2; 11 doubles it again. [R1 · Lab 1 · A]

31. A, B, D) Acknowledgement of delivered data / In-order delivery / Retransmission of lost packets. TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong. [R2 · 2.2 · U]

32. A) QUIC. QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes. [R2 · 2.2 · U]

33. B) On or near the machine (edge). Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it. [R2 · 2.3 · U]

34. C) The slowest small fraction of requests. For example the 99th-percentile delay; one slow server can delay a whole request. [R2 · 2.4 · U]

35. A) Putting computing close to the base station. Operator servers near the base station shorten the round trip. [R2 · 2.5 · U]

36. C) 0.81. V(3) = 1, V(2) = 0.9 × 1 = 0.9, V(1) = 0.9 × 0.9 = 0.81. [R3 · 3.2 · A]

37. D) 0.325. Target = 1 + 0.9 × 0.5 = 1.45; TD error = 1.25; new Q = 0.2 + 0.1 × 1.25 = 0.325. [R3 · 3.3 · A]

38. A, C) +speed every step / −energy used every step. A and C give feedback every step. B and D are sparse: they arrive only at the end. [R3 · 3.3 · U]

39. B) 0.77. 0.9 × 0.8 + 0.1 × (0.2 + 0.5 + 0.8) ÷ 3 = 0.72 + 0.05 = 0.77. Careful: if exploration chose only the OTHER two arms the answer would be 0.9 × 0.8 + 0.1 × 0.35 = 0.755. [R3 · 3.4 · A]

40. D) Training on many varied simulated conditions. If the policy works across many random variations, the real world looks like one more variation. [R3 · 3.5 · U]

41. A, C, D) LiDAR / Radar / Ultrasonic. LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth. [R4 · 4.1 · U]

42. A) 8.33 m. 100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts. [R4 · 4.2 · A]

43. B) 50.12 m. Weights $1/\sigma^2$: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates. [R4 · 4.3 · A]

44. C) 50 m. Z = f × B ÷ d = 1,000 × 0.3 ÷ 6 = 50 m. [R4 · 4.3 · A]

45. A) Level 2. Level 2 = partial automation; the human remains the driver. [R4 · 4.4 · U]

46. C) 112 GB. $7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split. [R5 · 5.2 · A]

47. D) 143 tokens/s. Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s. [R5 · 5.3 · A]

48. B) 24 hops. Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound. [R5 · 5.5 · A]

49. A, B, D) Fusing operations to cut memory traffic / Overlapping communication with computation / Using a larger batch per chip. A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls. [R5 · 5.4 · U]

50. D) 8×. Matrix multiply = $2n^3$ FLOPs; doubling n gives $2^3 = 8$×. For large sizes the measured time grows close to 8× too. [R5 · Lab 5 · A]

Part C · Open questions (sample answers, key points and marking guide)

Each open question is worth 5 marks. The key points below are a guide, not the only correct answer: give credit to any valid alternative that is correct and well reasoned.

51. Weights 512 × 2,048 = 1,048,576 → 2.1 MB; FLOPs per input 2,097,152. Batch 1 ≈ 1.0 FLOP/byte < 20 → memory-bound, ≈ 0.5 TFLOP/s (5%). Batch 32 ≈ 29.7 FLOP/byte > 20 → compute-bound, ≈ 10 TFLOP/s. Batching reuses each fetched weight for many inputs, raising FLOPs per byte. Marks: intensities (2), classification and TFLOP/s (1), explanation (2). Accept small differences from rounding or from ignoring activation bytes. [R1 · 1.4 · A]

52. Advantages: privacy (face data stays in the Secure Enclave), low latency, works offline, energy-efficient NPU. Limitations: limited compute and memory, unsupported operations fall back to the CPU, models must be small or quantised. Large chatbot: usually cloud or hybrid, because its weights exceed phone memory (or a small on-device model + cloud). Marks: 1 per valid point (4), justified decision (1). Accept other valid points, e.g. updates or battery life. [R1 · 1.5 · E]

53. Strong answers choose an on-premises edge GPU server or hybrid: data stays inside the hospital (privacy and law), LAN latency is low, X-ray images are large (bandwidth), the cloud can still be used for training or overflow. Cost: buying a server vs renting. Marks: recommendation (1), four justified factors (4). Accept a well-argued cloud answer that addresses privacy (encryption, anonymisation, contracts) and the 2-second budget. [R2 · 2.3 · E]

54. Traffic = 2 × 255/256 × 14 ≈ 27.9 GB; time ≈ 0.28 s; efficiency = 8.2 ÷ (8.2 + 0.28) ≈ 96.7%. Improvements: overlap communication with compute, larger batch per chip (more compute per byte), gradient compression, hierarchical collectives. Marks: 3 calculations, 2 for one improvement with a correct reason. [R2 · 2.4 · A]

55. Naive: +1 per item put in the bin → the robot tips the bin and re-collects. Better: reward the reduction of rubbish on the floor (or clean area), penalties for knocking things over and for energy. Dense if given every step, sparse if only when the room is clean. Comparison: the naive one rewards activity, the better one rewards the real goal. Marks: hack (2), improved reward (2), sparse/dense with comparison (1). Accept other plausible hacks. [R3 · 3.3 · C]

56. Failure: epsilon = 0 means no exploration; the agent exploits the first route that reached the goal, so unvisited squares are never updated and a shorter route is never found. Fix: epsilon about 0.1-0.2 (or decaying epsilon). Epsilon = 0.5: values are learned everywhere but the path looks messy and hits the −1 squares more often. Bandit link: the greedy agent found the best arm 0% of the time. ε-greedy explores at random at a fixed rate; UCB adds a bonus $\sqrt{(2 \ln t / n)}$ to little-tried actions and explores where it is uncertain. Marks: diagnosis (2), fix (1), ε = 0.5 prediction (1), comparison (1). Accept other reasonable fixes such as optimistic initial values. [R3 · Lab 3 · An]

57. 13.89 m/s; blind distance 2.08 m; braking $13.89^2 \div 14 \approx 13.78$ m; total ≈ 15.9 m. Detected at 15 m: 12.92 m left to brake, $v = \sqrt{(13.89^2 - 2 \times 7 \times 12.92)} \approx 3.5$ m/s at impact. Lesson: early, reliable detection (range) and low latency decide whether braking works; both must be in the design budget. Marks: 2 + 2 + 1. [R4 · 4.4 · A]

58. Speeds: 95, 250, 133, 67, 182 m/s. Mean t = 0.17 s → v ≈ 118 m/s (average the times, not the speeds). True delay is only 20 ÷ 343 ≈ 0.058 s, shorter than human reaction time (about 0.2 s), so reaction errors are as big as the signal. Averaging n trials shrinks random noise by about $\sqrt{n}$ (sensor fusion idea, 4.3) but cannot remove a systematic bias, e.g. reacting faster to sound than to sight. Improvements: a longer distance (100 m → 0.29 s), many more trials, or phyphox Acoustic Stopwatch on two phones (no human reaction). Marks: calculations (2), noise explanation (1), averaging vs bias (1), improvement (1). [R4 · Lab 4 · An]

59. $6 \times 7 \times 10^9 \times 1.4 \times 10^{12} = 5.88 \times 10^{22}$ FLOPs; $256 \times 200 \times 10^{12} \times 0.4 = 2.05 \times 10^{16}$ FLOP/s → ≈ 2.87×10^6 s ≈ 33.2 days; ≈ 204,167 chip-hours ≈ $408,333; 204,167 h × 0.4 kW × 1.1 ≈ 89.8 MWh. Raising MFU (fusion, better batching, overlap) cuts time, cost and energy in proportion. Marks: 4 values, 1 for the change with reason. Accept "more chips" only if the answer notes that cost and energy stay about the same. [R5 · 5.4 · A]

60. First call: tracing + XLA compilation. FLOPs = $2n^3$: 1,024 → 2.15 GFLOP ÷ 0.0021 s ≈ 1.0 TFLOP/s (≈ 13% MFU); 4,096 → 137 GFLOP ÷ 0.034 ≈ 4.0 TFLOP/s (≈ 50%); 8,192 → 1.10 TFLOP ÷ 0.27 ≈ 4.1 TFLOP/s (≈ 50%). 1,024 → 4,096 is 64× the FLOPs but only ≈ 16× the time, because at small sizes fixed overhead (launch, transfer) dominates and the GPU is not full. 4,096 → 8,192: 0.27 ÷ 0.034 ≈ 7.9 ≈ 8×, both sizes are compute-bound. Prediction for 16,384: about 8 × 0.27 ≈ 2.2 s, if memory allows. Marks: compilation (1), TFLOP/s and MFU (2), ratio explanation (1), prediction (1). Accept answers that also mention timing noise or GPU clock changes. [R5 · Lab 5 · P]